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Genome Note

First genome sequence of *Aeromonas hydrophilia* novel sequence type 658 strain isolated from livestock in South Africa

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ABSTRACT

Objectives: The underlying resistance mechanisms, defence systems, mobilome, virulome, clonality and global phylogenetic relationship of a novel sequence type (ST) 658 *Aeromonas hydrophilia* (A34a) isolated from a pig abattoir in South Africa was determined using whole-genome sequence (WGS) technology. **Methods:** Following isolation on chromogenic agar (CHROMID[®] CARBA SMART), microbial identification and antibiotic susceptibility testing were performed using a VITEK[®] 2 platform. Genotyping involved WGS performed with an Illumina MiSeq platform.

Results: The antibiotic resistome agreed with the resistance phenotype of the isolate and included antibiotic resistance determinants for β -lactams (*bla*_{CPHA3} and *bla*_{OXA-724}). BLASTn analysis of resistome-encoding contigs affirmed chromosomally-mediated resistance. BURST algorithmic analysis identified the novel ST658 as a satellite variant. Virulome analysis predicted virulence genes of *Aeromonas* whose expression are critical for establishing infection in the host. Global phylogenomic analyses showed strain A34a is closely related to two international isolates from Sri Lanka (Ae25) and the USA (RU34A), although there is little to suggest that it was imported from abroad.

Conclusion: This is the first report on the genomic analysis of a novel ST658 *A. hydrophilia*, offering useful insights into its pathogenicity and global phylogenetics.

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Aeromonas spp. is a non-spore-forming, oxidase-positive Gram-negative bacillus widely distributed in aquatic environments (freshwater, estuaries and marine), foods, domesticated pets, birds and natural soils. It is implicated in a wide spectrum of disease syndromes among warm and cold-blooded animals. Several species, such as *Aeromonas hydrophilia*, *Aeromonas veronii* biovar *sobria*, *Aeromonas schubertii*, *Aeromonas caviae* and *Aeromonas dhakensis*, have been implicated in diarrhoeal disease, wound infections including necrotizing fasciitis, and bacteraemia in humans [1,2]. Here we present the emergence of a novel sequence type (ST) 658 *A. hydrophilia* (A34a) isolated from a pig slaughterhouse in KwaZulu-Natal Province of South Africa.

The chromogenic screening medium CHROMID[®] CARBA SMART (bioMérieux, Marcy-l'Étoile, France) was used to isolate

strain A34a. Phenotypic microbial identification and antibiotic susceptibility testing were performed using a VITEK[®] 2 system (bioMérieux). Strain A34a was streaked onto nutrient agar (Sigma-Aldrich, St Louis, MO, USA) and was incubated at 37 °C for 24 h. Following incubation, a QIAamp[®] DNA Mini Kit (QIAGEN, Hilden, Germany) was used to extract the total genomic DNA of the isolate. A paired-end library was prepared using a Nextera XT DNA Sample Preparation Kit and then whole-genome sequencing (WGS) was carried out on an Illumina MiSeq system (Illumina Inc., San Diego, CA, USA). The sequenced reads were quality trimmed using Sickle v.1.33 (<https://github.com/najoshi/sickle>) and were de novo assembled using SKESA Assembler v.2.3 to obtain 91 contigs (95× coverage), the largest being 300 204 bp with an *N*₅₀ value of 60 834 bp. Default parameters were used for all software unless otherwise specified. All contiguous sequences were then submitted to GenBank, where gene annotation was implemented using the NCBI Prokaryotic Genome Annotation Pipeline (PGAP).

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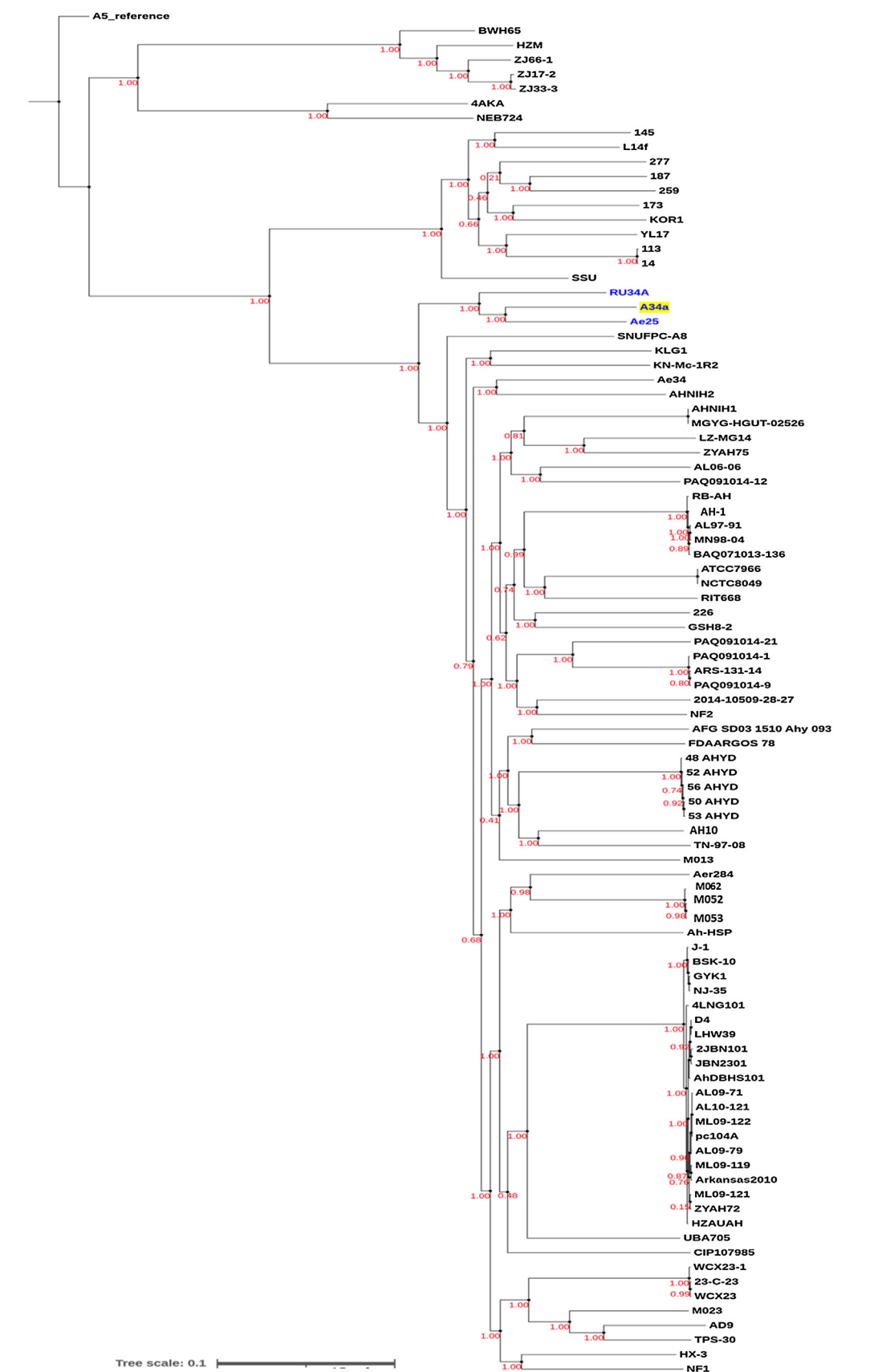


Fig. 1. Global phylogenomic tree showing the evolutionary relationship between *Aeromonas hydrophilia* A34a and other *A. hydrophilia* isolates of international origin. *Aeromonas veronii* A5 (accession no. **VVVTX000000000**) of the Aeromonadaceae family served as the outgroup or reference isolate and the branch labels refer to the bootstrap values. Strain A34a was in the same clade as Ae25 from Sri Lanka (fish, Koi carp) and RU34A from the USA (unknown source), with the same resistome profile (*bla*_{C_{PH}A3} and *bla*_{OXA-724}).

The genome features were as follows: genome size, 4 988 587 bp; GC content, 60.07%; total coding sequences (CDS), 4770; number of coding genes, 4672; number of RNA genes, 98; number of rRNAs, 2; and number of tRNAs, 88. The novel sequence type was defined as ST658 by the *Aeromonas* MLST database (<https://pubmlst.org/aeromonas/>). BURST (Based Upon Related Sequence Types) algorithmic analysis identified the novel ST658 as a satellite variant (more distantly related) with no known parent ancestry with global curated *Aeromonas* sequence types. CRISPRCasFinder (<https://crisprcas.i2bc.paris-saclay.fr/CrisprCasFinder/Index>) using default settings identified one confirmed CRISPR element in the genome responsible for adaptive immunity mechanisms to protect the strain against invading viral attacks [3].

The antibiotic resistome agreed with the resistance phenotype of the isolate and included antibiotic resistance determinants for β -lactams (*bla*_{C_{PHA3} and *bla*_{OXA-724}) (according to ResFinder v.3.1). Mobilome analysis indicated that the isolate lacked plasmids, integrons and intact prophages. Moreover, BLASTn analysis to elucidate the genetic environment of resistome-encoding contigs revealed that they were located on the chromosome with >97% homology to *A. hydrophila* isolate MX16A (accession no. **CP018201.1**). This affirmed the chromosomally-mediated β -lactam resistance reported in *Aeromonas* spp. [4] and the absence of mobile genetic support in the isolate.}

Whole virulome analysis using the pathogenic bacteria database, VFAnalyzer (<http://www.mgc.ac.cn/cgi-bin/VFs/v5/main.cgi?func=VFAnalyzer>), predicted a total of 168 virulence-encoding genes belonging to seven major virulence factor classes of *Aeromonas*, namely adherence factor (Flp type IV pili, mannose-sensitive haemagglutinin pilus, polar flagella, tap type IV pili and type I fimbriae), secretion system (T2SS and T6SS), toxin (enterotoxin and haemolysin), anaerobic respiration (nitrate reductase), antiphagocytosis (capsular polysaccharide), immune evasion (capsule and LOS) and iron uptake. Differential expression of the accrued virulome confers a competitive advantage, contributing to its success as a pathogen, and is critical for establishing host infection [5].

The genetic relatedness of the isolate with other global isolates is represented in a phylogenomic tree drawn with CSI Phylogeny 1.4 (<https://cge.cbs.dtu.dk/services/CSIPhylogeny/>), which generates the core genome alignment, identifies single nucleotide polymorphisms (SNPs) from WGS data, and filters and validates the SNP positions. *Aeromonas veronii* A5 (accession no. **VWTX00000000**) of the Aeromonadaceae family served as the reference isolate. The recovered core genome alignment from CSI Phylogeny tool was then used to infer phylogeny using IQ-TREE (<http://www.iqtree.org/>) and the generated tree was edited with Figtree (Fig. 1). The results demonstrated *A. hydrophila* to be a diverse species with many unresolved clades (Fig. 1). Strain A34a is closely related to two isolates, namely Ae25 from Sri Lanka (fish, Koi carp) and RU34A from the USA (unknown source), with the same resistome profile (*bla*_{C_{PHA3} and *bla*_{OXA-724}), although there is little to suggest that it was imported from abroad. The close relationship between these international isolates with this novel strain shows the diversity, evolution and dissemination of *A. hydrophila* worldwide.}

The results of this bioinformatic analysis offer useful insights into the pathogenomics and the global phylogenomic relationship of the novel *A. hydrophila* strain cultured from a pig abattoir in South Africa.

Nucleotide sequence accession no.

This Whole Genome Shotgun project has been deposited at DDBJ/ENA/GenBank under BioProject **PRJNA564235** with the accession no. **VWTU00000000**. The version described in this paper is version VWTU10000000.

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Competing interest

None declared.

Ethical approval

This study was approved by the Animal Research Ethics Committee [approval no. AREC/079/018D], University of KwaZulu-Natal (UKZN). Permission for animal sampling was obtained from the Department of Agriculture, Forestry and Fisheries, Republic of South Africa [approval no. Ref 12/11/15].

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